LUIGUI GALLARDO-BECERRA

BIOINFORMATICIAN | MOLECULAR BIOLOGIST

+1 619 602 0725 | luiguimichelgallardo@gmail.com | www.lmgb.xyz | github: LuiguiGallardo | linkedin: luiguigallardo

CAREER SUMMARY

Computational Biologist and Data Scientist with 6+ years of expertise managing high-volume NGS datasets across genomics, transcriptomics, and microbiome research. Proven ability to build robust, scalable bioinformatics pipelines and apply machine learning algorithms to complex biological data. Experienced in high-performance computing with a strong scientific track record of 15 peer-reviewed publications.

EXPERIENCE

Data Specialist | TEKSystems @ Meta **July 2026 - Present**

- **Content Classification:** Performed systematic classification and annotation of digital content to improve Content Brand Suitability standards across Meta's advertising ecosystem.
- **Content Labeling:** Evaluated and labeled content at scale, applying established guidelines to ensure brand-safe environments for advertisers.
- **Data Quality:** Contributed to data quality initiatives that feed into machine learning models for automated content moderation and suitability scoring.
- **Policy Refinement:** Collaborated with cross-functional teams to identify edge cases and provide feedback that refines classification frameworks and policy guidelines

STEM Python Expert (Chemistry and Biology) | Mercor **April 2026 - Present**

- **LLM Benchmarking:** Designed and solved complex coding challenges to benchmark Large Language Models (LLMs) and rigorously verify their logical reasoning.
- **Code Evaluation:** Evaluated and debugged AI-generated Python code across STEM domains to ensure strict accuracy, optimal performance, and adherence to best practices.
- **RLHF & Dataset Generation:** Annotated, ranked, and corrected model responses to generate high-quality datasets for AI training via reinforcement learning from human feedback (RLHF).

Research Assistant (Bioinformatics & Data Science) | Institute of Biotechnology, UNAM **January 2019 - July 2025**

- **Bioinformatics workflow engineering:** Designed and implemented automated, reproducible pipelines using Snakemake, Nextflow, Python, and Bash for large-scale NGS data processing.
- **NGS data analysis:** Processed and analyzed whole-genome sequencing, RNA-seq, 16S rRNA, metagenomic, and metatranscriptomic datasets, including QC, alignment, quantification, and downstream interpretation.
- **Machine learning and statistical modeling:** Applied supervised and unsupervised learning methods to identify biomarkers, classify samples, and uncover structure in high-dimensional omics datasets.
- **High-performance computing (HPC):** Administered HPC clusters, optimized pipeline performance, and managed large-scale computational workloads across Linux/SLURM environments.
- **Scientific software development:** Created custom tools and analysis scripts in Python, R, and Perl to address project-specific computational challenges and automate complex workflows.

Data Engineer | Appen **October 2016 - December 2018**

- **ETL Engineering:** Developed automated PostgreSQL ETL pipelines to efficiently integrate, clean, and transform high-volume datasets.
- **Data Quality Control:** Implemented robust QC workflows for machine learning models, ensuring high reliability and reproducibility in downstream analyses.
- **Data Collection:** Automated data collection and validation systems to minimize manual workload and streamline cross-project reporting.

KEY SKILLS

Programming & Scripting: Python, R, Bash, SQL, Perl, C#, PostgreSQL, JavaScript

Bioinformatics Tools & Frameworks: Snakemake, Nextflow, Galaxy, QIIME2, Bioconductor, Biopython, Samtools, BLAST, BWA, Bowtie2

NGS Data Analysis: Illumina/Nanopore workflows; quality control (FastQC, MultiQC); read alignment; variant calling; de novo assembly; genome/transcriptome annotation

Microbiome & Multi-Omics Analysis: Metagenomics, metatranscriptomics, 16S rRNA profiling, comparative genomics, phylogenetics, transcriptomics

Data Engineering & Management: SQL/NoSQL database design, ETL workflows, data warehousing, Git version control

Computing & Infrastructure: Linux/Unix, HPC cluster administration, cloud computing (AWS, Google Cloud), containerization (Docker, Singularity), Conda environments

Data Science & Visualization: Statistical modeling, machine learning, visualization with ggplot2, seaborn, matplotlib, and R Shiny

Professional Skills: Project management, scientific communication, cross-functional collaboration, technical writing, problem-solving

Languages: English (Full professional proficiency), Spanish (Native)

Education

Ph.D. in Biochemistry | National Autonomous University of Mexico (UNAM) | Mexico January 2019 - July 2025
Master of Science in Biochemistry | National Autonomous University of Mexico (UNAM) | Mexico August 2016 - January 2019
Bachelor of Science in Biology | University of Guadalajara (UDG) | Guadalajara, Mexico August 2012 - January 2016

Selected Publications (15 total)

Gallardo-Becerra L, Cornejo-Granados F, Bikel S, Arenas I, López-Leal G, Alvarado-Gonzalez C, et al. Bioactive plasmid- and phage-encoded antimicrobial peptides (AMPs) in the human gut: a metatranscriptome–virome profiling reveals exploratory links to metabolic human diseases. Microb Ecol. 2025 Nov 28.

Gallardo-Becerra L, Cornejo-Granados F, Romero-Hidalgo S, Lopez-Zavala AA, Cota-Huízar A, Cervantes-Echeverría M, et al. Host genome drives the microbiota enrichment of beneficial microbes in shrimp: exploring the hologenome perspective. Anim Microbiome. 2025 May 22;7(1):50.

Gallardo-Becerra L, Cervantes-Echeverría M, Cornejo-Granados F, Vazquez-Morado LE, Ochoa-Leyva A. Perspectives in searching antimicrobial peptides (AMPs) produced by the microbiota. Microb Ecol. 2023 Dec 1;87(1):8.

Gallardo-Becerra L, Cornejo-Granados F, García-López R, Valdez-Lara A, Bikel S, Canizales-Quinteros S, et al. Metatranscriptomic analysis to define the Secrebiome, and 16S rRNA profiling of the gut microbiome in obesity and metabolic syndrome of Mexican children. Microb Cell Fact. 2020 Dec;19(1):61.

Gallardo-Becerra L, Cornejo-Granados F, Leonardo-Reza M, Ochoa-Romo JP, Ochoa-Leyva A. A meta-analysis reveals the environmental and host factors shaping the structure and function of the shrimp microbiota. PeerJ. 2018 Aug 10;6:e5382.